T123-PS-106 Design Document (Week 1)

Apartment Rental Agreement, Escalation, Deposit Deduction and Move-out Settlement Engine

Team T123 · Yashwanth K S (1AT24IS224) · Yogish Gowda S (1AT24IS225) · ISE D · Language: Python 3.10+

1. Overview

The engine is a command-file driven program (no UI). It reads lease, payment and event files, then prints monthly invoices, rent ledgers, lease state and the move-out settlement statement. All money is handled as `Decimal` and rounded half-up to whole rupees.

2. Glossary

Term	Meaning
Lease	Agreement between landlord and tenant with start date, term, rent, deposit, lock-in, notice period and escalation clause
Base rent	Monthly rent agreed in the lease, before maintenance. Escalation applies only to this
Maintenance	Society charge passed through to the tenant at actuals with zero markup
Security deposit	Amount (2 to 10 months' rent) held by the landlord and used for settlement
Lock-in	Initial period during which the tenant cannot leave without forfeiting part of the deposit
Notice period	Advance notice the tenant must give before vacating (30 or 60 days per lease)
Escalation	Percentage increase of base rent on each 12-month anniversary of a renewed lease
Pro-rating	Charging a partial month by actual days occupied divided by days in that month
Due date	The 5th of each month

Term	Meaning
Grace period	Up to and including the 10th, with no late charge
Late fee	Flat Rs 500, charged once per invoice from the 11th
Late interest	2% per month simple interest on overdue rent, computed daily
DEFAULT	Flag raised when any invoice is unpaid for 60 or more days
NEGOTIATED	Flag allowing a renewal rent increase above 10%
UNCAPPED	Flag removing the 10% maintenance cap
Forfeiture	Part of the deposit kept by the landlord for breaking the lock-in
Notice shortfall	Rent charged for the days by which the tenant's notice fell short
Inspection	Move-out check producing itemised deductions from the rate card
Normal wear	Ordinary ageing of the flat, never deducted
Settlement	Final statement of deposit minus deductions plus credits, giving the refund (or amount due)
Vacation date	Date the tenant hands back the flat

3. Business-rule register

Lease terms and billing

ID	Rule
BR-01	A standard lease term is 11 months unless the lease specifies otherwise
BR-02	Security deposit must be between 2 and 10 months' base rent; otherwise lease creation is rejected
BR-03	Rent is due on the 5th of each month

ID	Rule
BR-04	First month rent = base rent × days occupied ÷ days in that month, rounded to the nearest rupee
BR-05	Last month rent uses the same formula with actual days up to the lease end date
BR-06	Days in month are always the actual days of the calendar month in question (28 to 31)

Escalation

ID	Rule
BR-07	Base rent escalates by 5% on each 12-month anniversary of a renewed lease
BR-08	If the agreement specifies a different escalation percentage, it overrides 5%
BR-09	Escalation applies to base rent only, never to maintenance
BR-10	No escalation applies within the first 11 months

Late payment and payments

ID	Rule
BR-11	Payment up to and including the 10th is within grace; no late fee or interest
BR-12	From the 11th, a flat Rs 500 late fee applies, charged once per invoice
BR-13	From the 11th, interest accrues at 2% per month simple on overdue rent, computed per day
BR-14	A payment is applied in this order: interest, then late fee, then oldest unpaid rent
BR-15	A payment that covers only interest and late fee leaves rent outstanding unchanged

ID	Rule
BR-16	An invoice unpaid for 60 or more days raises a DEFAULT flag
BR-17	A lease with a DEFAULT flag cannot be renewed
BR-18	A DEFAULT flag triggers a termination notice

Lock-in

ID	Rule
BR-19	Breaking the lease within the lock-in period forfeits 1 month's rent from the deposit
BR-20	The forfeiture is 2 months' rent if lock-in is 6 months or more and the break is within the first 3 months
BR-21	Leaving after the lock-in period ends carries no forfeiture

Notice

ID	Rule
BR-22	Tenant notice period is as per the lease (30 or 60 days)
BR-23	Any shortfall in tenant notice is charged as rent for the missing days
BR-24	Landlord notice must be at least 60 days; shorter landlord notice is rejected
BR-25	Valid landlord notice waives any lock-in forfeiture

Maintenance

ID	Rule
BR-26	Maintenance is passed through at actuals with zero markup

ID	Rule
BR-27	Maintenance for a partial month is pro-rated by days
BR-28	Monthly maintenance is capped at 10% of rent unless the lease has the UNCAPPED flag

Move-out inspection

ID	Rule
BR-29	Painting is Rs 15 per sq ft, charged once, if the stay is under 24 months
BR-30	If the stay is 24 months or more, painting is charged at 50% (Rs 7.50 per sq ft)
BR-31	Deep cleaning is a flat Rs 3,000
BR-32	Itemised damages are charged at their listed amounts
BR-33	Normal wear items are excluded from deductions
BR-34	Total inspection deductions cannot exceed deposit plus 1 month's rent

Deposit interest and refund timing

ID	Rule
BR-35	If deposit exceeds 3 months' rent, the landlord pays 6% p.a. simple interest on the excess for the tenancy duration, credited at settlement
BR-36	If deposit is 3 months' rent or less, no deposit interest is paid
BR-37	The refund is due within 15 days of the vacation date
BR-38	If the settlement date is later than 15 days, add 1% per month interest on the net refundable amount in favour of the tenant

Renewal

ID	Rule
BR-39	Renewal is offered 45 days before lease expiry
BR-40	Accepting a renewal resets lock-in to 0 unless the renewal specifies otherwise
BR-41	A renewal with rent increase above 10% requires the NEGOTIATED flag, otherwise it is rejected
BR-42	A renewal with an increase of exactly 10% is accepted without the flag

Settlement

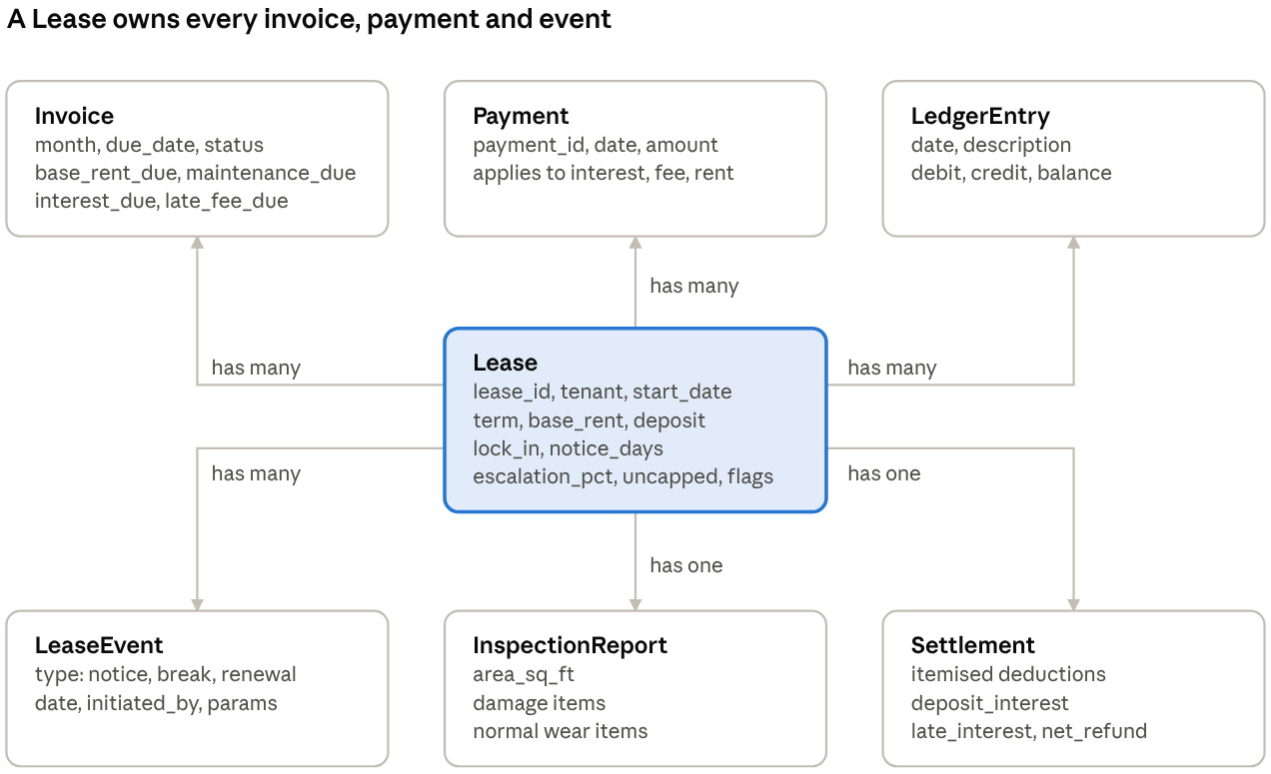
ID	Rule
BR-43	Settlement applies charges in this order: unpaid rent and interest, notice shortfall, lock-in forfeiture, inspection deductions (capped), deposit interest credit, late-refund interest
BR-44	All computed amounts are rounded half-up to whole rupees

4. Edge cases mapped to rules

ID	Edge case	Expected handling	Rules
EC-01	Lease starts on the 31st and next month has 30 days	Day counts use each actual month; no invalid date is created	BR-04, BR-06
EC-02	Payment received exactly on the 10th	Treated as within grace; no late fee	BR-11
EC-03	Break at exactly 3 months into a 6-month lock-in	Counts as within the first 3 months, so 2 months forfeited	BR-20
EC-04	Notice given 29 days before vacate date	Shortfall charged against the lease notice period	BR-22, BR-23

ID	Edge case	Expected handling	Rules
EC-05	Deposit exactly equal to 3 months' rent	Excess is zero, so no interest	BR-35, BR-36
EC-06	Inspection deductions exceed deposit plus 1 month's rent	Deductions capped at deposit plus 1 month's rent	BR-34
EC-07	Renewal with exactly 10% increase	Accepted without NEGOTIATED flag	BR-41, BR-42
EC-08	Partial payment covers interest and late fee but zero rent	Rent balance unchanged; only interest and late fee cleared	BR-14, BR-15

5. Domain model (class diagram)



class diagram: Lease and its 6 related classes

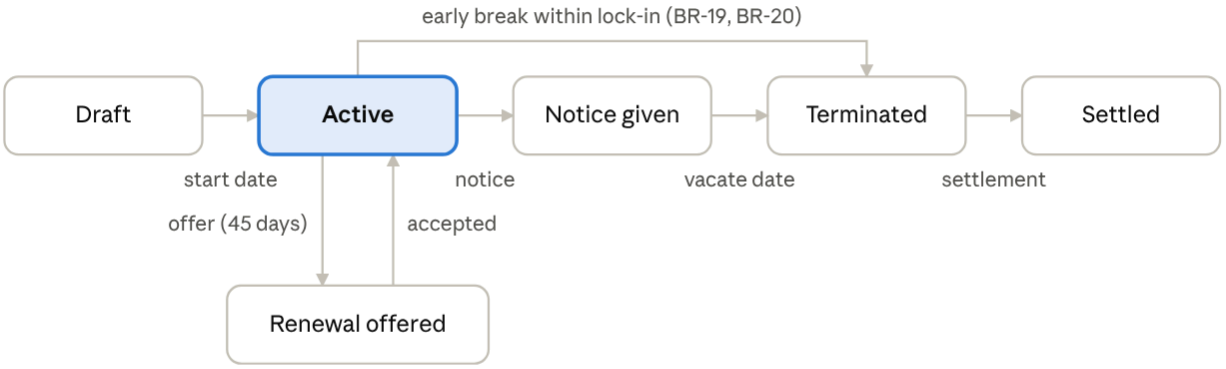
The Lease is the central entity; every other class belongs to exactly one Lease.

Class	Key attributes	Relationships
Lease	lease_id, tenant, start_date, term_months, base_rent, deposit, lock_in_months, notice_days, escalation_pct, uncapped, status, flags	has many Invoice, Payment, LeaseEvent; has one InspectionReport and one Settlement
Invoice	invoice_id, month, base_rent_due, maintenance_due, interest_due, late_fee_due, paid_amounts, due_date, status	belongs to Lease; receives PaymentAllocation
Payment	payment_id, date, amount	belongs to Lease; splits into PaymentAllocation
PaymentAllocation	component (interest, late fee, rent), amount, invoice_id	links Payment to Invoice
LedgerEntry	date, description, debit, credit, balance	belongs to Lease
LeaseEvent	type (NOTICE, BREAK, RENEWAL_OFFER, RENEWAL_ACCEPT, EXTENSION), date, initiated_by, params	belongs to Lease
InspectionReport	area_sq_ft, items (damages), normal_wear_items	belongs to Lease
Settlement	deductions (itemised), deposit_interest, late_refund_interest, net_refund, settlement_date	belongs to Lease
RateCard	painting_rate, painting_discount_after_24m, deep_cleaning, damage_amounts	used by SettlementService

Engine classes: CommandParser (reads files), BillingService (invoices, escalation, maintenance), PaymentService (allocation, late fee, interest, DEFAULT), EventProcessor (notice, break, renewal), SettlementService (move-out statement), Printer (lease state, ledger, settlement).

6. State diagram for the Lease

A lease ends in Settled; renewal loops back to Active



DEFAULT flag on Active blocks renewal and triggers a termination notice (BR-16 to BR-18)

lease state diagram: 6 states

The table below lists every transition with its rule IDs.

From	Event	To	Rules
(new)	create lease	DRAFT	BR-01, BR-02
DRAFT	start date reached	ACTIVE	BR-03
ACTIVE	renewal offered 45 days before expiry	RENEWAL_OFFERED	BR-39
RENEWAL_OFFERED	accepted and valid	ACTIVE (renewed)	BR-40, BR-41, BR-42
RENEWAL_OFFERED	rejected or DEFAULT	ACTIVE or TERMINATED at expiry	BR-17, BR-41
ACTIVE	invoice unpaid 60+ days	ACTIVE with DEFAULT flag	BR-16
ACTIVE (DEFAULT)	termination notice raised	NOTICE_GIVEN	BR-18
ACTIVE	tenant or landlord notice	NOTICE_GIVEN	BR-22, BR-24
ACTIVE	early break in lock-in	TERMINATED	BR-19, BR-20
NOTICE_GIVEN	vacation date reached	TERMINATED	BR-23

From	Event	To	Rules
TERMINATED	inspection and settlement computed	SETTLED	BR-43

7. Design assumptions

ID	Topic	Decision
A-01	Lock-in break window	"Within the first 3 months" includes the date exactly 3 months after the start date (EC-03)
A-02	Late interest	Overdue rent × 2% × days overdue ÷ 30
A-03	Deposit interest	Excess deposit × 6% × actual tenancy days ÷ 365
A-04	DEFAULT timing	The 60 days are counted from the invoice due date (the 5th)
A-05	Maintenance cap	10% of full monthly rent, then pro-rated for partial months
A-06	Notice shortfall rate	Monthly base rent ÷ actual days in the vacation month, per missing day
A-07	Deduction cap	The cap of deposit plus 1 month's rent applies to inspection deductions only (BR-34)
A-08	Rounding	Each computed charge is rounded half-up to whole rupees (BR-44)

Where the brief is open to interpretation, these assumptions are recorded in the decision log and will be revised if the evaluators clarify a rule.